Mathematical and Computational Methods

Lecture 7: Differentiation I: limits and the rules

With the algebra and geometry of the module's first half behind us, we change subject to the central tool of applied mathematics: the *derivative*. This lecture builds it from the ground up. We start with the *limit*—the idea of a value approached but not necessarily reached—use it to pin down what it means for a function to be *continuous*, and then to define the derivative, which we learn to read three ways at once: geometrically as the slope of a tangent, physically as a rate of change, and—for the data-science students—as a *sensitivity*. We then assemble the standard derivatives and the *rules* (sum, product, quotient and chain) that let us differentiate almost any function by hand, before finishing with three techniques—parametric, implicit and logarithmic differentiation—for functions that are not handed to us as a tidy $y = f(x)$.

Where this fits. The raw materials are all from Lecture 1: the elementary functions, their domains, composition (which becomes the chain rule), and monotonicity. Differentiation is the engine of Lecture 8, where we use it to find maxima and minima, sketch curves and solve optimisation problems, and it is the exact reverse of the integration developed in Lectures 9–10. The standard derivatives and rules collected here are a reference point you will return to for the rest of the course and beyond.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Evaluate one-sided, infinite and limits at infinity, and locate asymptotes.
- Define continuity and explain why differentiability implies continuity.
- Differentiate from first principles and read the derivative three ways.
- Apply the standard derivatives and the sum, product, quotient and chain rules.
- Differentiate parametrically, implicitly and logarithmically.

1 Limits, continuity and the derivative

1.1 Limits at a point

Calculus rests on a single idea: what value does a function *approach* as its input approaches some target? This is captured by the limit.

Definition 1.1 (Limit at a point). We write

$$\lim_{x \to a} f(x) = L$$

to mean that $f(x)$ can be made as close as we like to the number L by taking x sufficiently close to a (but not equal to a). The value $f(a)$ itself—which may not even be defined—plays no role. (This intuitive statement is made fully precise by the ε–δ definition in the Calculus module.)

The parenthetical "but not equal to a" is the whole point: a limit probes the behaviour of f *near* a, not *at* a. This is exactly what lets us divide by a quantity that is heading to zero, as the next example shows.

Example 1.1 ***(A limit of the*** $\frac{0}{0}$ ***type).*** The function $f(x) = \dfrac{x^2-1}{x-1}$ is undefined at $x = 1$, where both numerator and denominator vanish. For every *other* x, though, we may cancel:

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1 \qquad (x \neq 1).$$

As $x \to 1$ the right-hand side approaches 2, so $\lim_{x \to 1} \dfrac{x^2-1}{x-1} = 2$, even though $f(1)$ does not exist.

For the functions in this course the everyday rules hold: limits respect sums, products and quotients (the quotient rule provided the denominator's limit is not 0), so a limit can usually be assembled from simpler ones.

1.2 One-sided limits, infinite limits, and limits at infinity

Three further kinds of limiting behaviour complete the picture, and each has a geometric meaning we will need.

One-sided limits

A function may approach different values from the two sides of a. We write

$$\lim_{x\to a^-} f(x) \quad (x \text{ approaching } a \text{ from below, } x < a), \qquad \lim_{x\to a^+} f(x) \quad (\text{from above, } x > a).$$

The two-sided limit $\lim_{x\to a} f(x)$ exists *precisely when* both one-sided limits exist and are equal.

Example 1.2 ***(A jump: the Heaviside step).*** For the step function $H(x)$ of Lecture 1 ($H = 0$ for $x < 0$, $H = 1$ for $x \geq 0$),

$$\lim_{x\to 0^-} H(x) = 0, \qquad \lim_{x\to 0^+} H(x) = 1.$$

The one-sided limits disagree, so $\lim_{x\to 0} H(x)$ does not exist: the graph jumps at 0.

Infinite limits

A function may grow without bound near a point. We write $\lim_{x\to a} f(x) = \infty$ to mean $f(x)$ increases beyond every bound as $x \to a$; this is not an existing (finite) limit but a convenient record of *how* the limit fails to exist, and it signals a *vertical asymptote* $x = a$. For example $\dfrac{1}{x^2} \to \infty$ as $x \to 0$ from either side, while $\dfrac{1}{x}$ behaves differently on the two sides,

$$\lim_{x\to 0^+} \frac{1}{x} = \infty, \qquad \lim_{x\to 0^-} \frac{1}{x} = -\infty,$$

just as $\tan x$ races off to $\pm\infty$ at $x = \frac{\pi}{2}$ (Lecture 1).

Limits at infinity

We also ask how a function behaves as x grows without bound. The notation $\lim_{x\to\infty} f(x) = L$ means $f(x)$ approaches L as x grows beyond every bound, and corresponds to a *horizontal asymptote* $y = L$ of the graph.

Example 1.3 ***(A limit at infinity).*** For $f(x) = \dfrac{3x^2 - 1}{x^2 + 5}$, divide top and bottom by the highest power x^2:

$$\frac{3x^2 - 1}{x^2 + 5} = \frac{3 - \frac{1}{x^2}}{1 + \frac{5}{x^2}} \xrightarrow[x\to\infty]{} \frac{3 - 0}{1 + 0} = 3.$$

The line $y = 3$ is therefore a horizontal asymptote, approached as $x \to -\infty$ too.

Note. A standard limit worth memorising is $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$ (with x in radians). It says $\sin x \approx x$ for small x, and it is the limit that ultimately produces the derivative of $\sin x$.

1.3 Continuity

The most convenient functions are those with no jumps or breaks, so that the limit at a point is simply the value there. This is continuity.

Definition 1.2 (Continuous function). A function f is *continuous at* a point a of its domain if

$$\lim_{x\to a} f(x) = f(a),$$

that is, the limit exists and equals the function's value. f is *continuous on an interval* if it is continuous at every point of that interval.

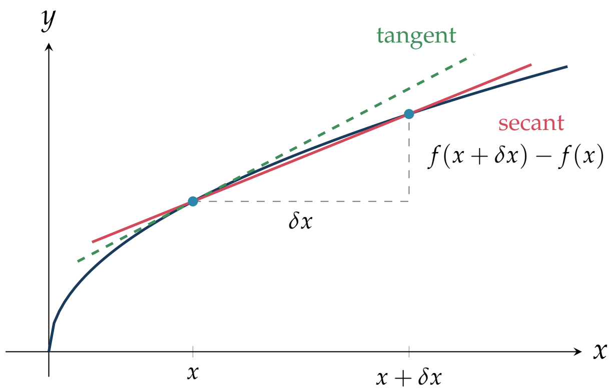
Three things must hold at once: $f(a)$ is defined, the limit exists, and the two agree. Informally, a continuous function is one whose graph can be drawn without lifting the pen: a jump (like the step function above) breaks continuity, while at a vertical asymptote the function is undefined, so continuity does not even apply there. The functions we use constantly—polynomials, e^x, $\ln x$, $\sin x$, $\cos x$, and their sums, products, quotients and compositions—are all continuous on their domains, which is why their limits are found simply by substituting.

1.4 The derivative

Fix a point on the graph of f and a nearby point a horizontal distance δx away. The straight line through the two points—a *secant*—has slope

$$\frac{f(x+\delta x) - f(x)}{\delta x},$$

the change in height divided by the change in x. As the second point slides towards the first ($\delta x \to 0$), the secant pivots into the *tangent* line, and its slope tends to the slope of the curve at x. That limiting slope is the derivative.



Definition 1.3 (The derivative). The *derivative* of f at x is

$$f'(x) = \lim_{\delta x\to 0} \frac{f(x+\delta x) - f(x)}{\delta x},$$

provided the limit exists, in which case f is *differentiable* at x. Its value is the slope of the tangent to $y = f(x)$ at the point $\big(x, f(x)\big)$.

Computing a derivative straight from this definition is called working *from first principles*. We do it once, to see the machinery, and then never again—the rules of Section 2 make it unnecessary.

Example 1.4 ***(The derivative of x^2 from first principles).*** With $f(x) = x^2$,

$$\frac{f(x+\delta x) - f(x)}{\delta x} = \frac{(x+\delta x)^2 - x^2}{\delta x} = \frac{2x\,\delta x + (\delta x)^2}{\delta x} = 2x + \delta x.$$

Letting $\delta x \to 0$ leaves $f'(x) = 2x$.

1.5 Three readings of the derivative

The single number $f'(x)$ carries three complementary meanings, and fluency in calculus means hearing all three at once.

- **Geometric — a slope.** $f'(x)$ is the slope of the tangent to the curve $y = f(x)$, measuring how steeply the graph rises or falls.
- **Physical — a rate of change.** If a quantity depends on time, its derivative is the instantaneous rate at which it changes: the derivative of position is velocity, and differentiating once more gives acceleration.
- **Sensitivity — a response to a nudge.** $f'(x)$ measures how strongly the output responds to a small change in the input: a large $|f'(x)|$ means a tiny nudge in x produces a big change in f. In data science this is decisive—if a model's error L depends on a parameter w, then $\frac{\mathrm{d}L}{\mathrm{d}w}$ says which way, and how hard, to adjust w to reduce the error. That is precisely the information the gradient-descent method of Lecture 8 exploits.

The first two readings are illustrated below; the third returns, made computational, in Lecture 8.

Example 1.5 ***(Equation of a tangent line).*** The tangent line at a point is the line through that point with slope f'. For $y = \sqrt{x}$ at $x = 4$ we have $f(4) = 2$ and $f'(x) = \dfrac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$. The point–slope form gives

$$y - 2 = \tfrac{1}{4}(x-4) \qquad \Longrightarrow \qquad y = \tfrac{1}{4}x + 1.$$

(The systematic treatment of tangents—and of the perpendicular *normal* lines—comes in Lecture 8.)

Example 1.6 ***(Velocity and acceleration).*** A particle moves along a line with displacement $s(t) = t^3 - 6t^2 + 9t$ metres after t seconds. Its velocity and acceleration are the first two derivatives,

$$v(t) = \dot{s} = 3t^2 - 12t + 9 \ (\mathrm{m\,s^{-1}}), \qquad a(t) = \ddot{s} = 6t - 12 \ (\mathrm{m\,s^{-2}}).$$

The particle is momentarily at rest when $v = 0$, i.e. $3(t-1)(t-3) = 0$, at $t = 1$ and $t = 3\,\mathrm{s}$. At $t = 1$ the acceleration is $a(1) = -6\,\mathrm{m\,s^{-2}}$ (it is about to move backwards), while at $t = 3$ it is $a(3) = +6\,\mathrm{m\,s^{-2}}$. The acceleration here is a derivative of a derivative—a *higher-order* derivative, which we define next.

1.6 Notation, higher derivatives, and a function that fails

Notation

Two notations are used side by side throughout the sciences, and you must be fluent in both.

Note (Newton and Leibniz notation). For $y = f(x)$ the derivative is written

$$f'(x) \quad \text{or} \quad \frac{\mathrm{d}y}{\mathrm{d}x},$$

the first due to Newton (compact; good for naming a function), the second to Leibniz (displays the variables; behaves like a ratio under the chain rule). A dot, $\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$, is reserved for differentiation with respect to *time*. To indicate the value of the derivative at a particular point we write $f'(a)$ or $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=a}$.

Higher-order derivatives

The derivative f' is itself a function, so it too can be differentiated. The result is the *second derivative* f'', and repeating gives derivatives of every order.

Definition 1.4 (Higher derivatives). The second derivative is $f''(x) = \dfrac{\mathrm{d}}{\mathrm{d}x} f'(x)$, written $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$ in Leibniz notation (or $\ddot{x}$ for a time derivative). In general the nth derivative is obtained by differentiating n times,

$$f^{(n)}(x) = \frac{\mathrm{d}^n y}{\mathrm{d}x^n} = \frac{\mathrm{d}}{\mathrm{d}x} f^{(n-1)}(x).$$

The second derivative measures how the *slope* is changing: in mechanics it is acceleration, and in Lecture 8 its sign will distinguish a peak from a trough and tell us which way a curve bends.

Example 1.7 ***(Successive derivatives).*** For $y = x^3$ we get $\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2$, then $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = 6x$, $\frac{\mathrm{d}^3 y}{\mathrm{d}x^3} = 6$, and every further derivative is 0. For the simple-harmonic displacement $x(t) = A\cos\omega t$, two differentiations give $\dot{x} = -A\omega \sin\omega t$ and $\ddot{x} = -A\omega^2 \cos\omega t = -\omega^2 x$, the equation of motion of an oscillator.

A function that fails to be differentiable

Differentiability is demanding: the limit defining $f'(x)$ may simply fail to settle on one value. The standard example is a graph with a sharp corner.

Example 1.8 ***(A corner: $f(x) = |x|$).*** For $x > 0$ the graph of $|x|$ is the line $y = x$ with slope $+1$; for $x < 0$ it is $y = -x$ with slope -1. The one-sided limits of the difference quotient at 0 are therefore

$$\lim_{\delta x \to 0^+} \frac{|\delta x| - 0}{\delta x} = +1, \qquad \lim_{\delta x \to 0^-} \frac{|\delta x| - 0}{\delta x} = -1.$$

They disagree, so the two-sided limit does not exist: $|x|$ is *not* differentiable at 0, even though it is perfectly continuous there.

Remark (ReLU, the workhorse activation). The same corner appears in the *rectified linear unit* $\mathrm{ReLU}(x) = \max(0, x) = \frac{1}{2}(x + |x|)$ from Lecture 1, the most widely used activation function in neural networks. It has slope 0 for $x < 0$ and slope 1 for $x > 0$, so by the identical one-sided argument it is differentiable everywhere *except* at 0, where the left and right slopes (0 and 1) disagree. In the gradient-descent training of Lecture 8 this is no obstacle: the derivative at the corner is simply fixed by convention (usually 0), since a single point rarely affects the descent.

This last point—continuous but not differentiable—is no accident, and it fixes the exact relationship between the two ideas.

Theorem 1.1 (Differentiable $\Rightarrow$ continuous). If f is differentiable at a, then f is continuous at a.

Proof. For $x \neq a$ write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}\,(x - a).$$

As $x \to a$ the first factor tends to $f'(a)$ (which exists, by hypothesis) and the second tends to 0, so $f(x) - f(a) \to f'(a) \cdot 0 = 0$. Hence $\lim_{x\to a} f(x) = f(a)$, which is exactly continuity at a. ■

Remark. The converse is false: $|x|$ is continuous at 0 but not differentiable there, so differentiability is *strictly stronger* than continuity. One useful direction does always hold—a differentiable function is automatically continuous—and we quietly used precisely this fact in the proof of the product rule below.

2 Standard derivatives and the rules

2.1 The standard derivatives

A handful of derivatives, each provable from first principles, serve as the building blocks for everything else. They are collected here for reference; throughout, n denotes a constant.

$f(x)$	x^n	e^x	$\ln x$	$\sin x$	$\cos x$	$\tan x$
$f'(x)$	$n\,x^{n-1}$	e^x	$\frac{1}{x}$	$\cos x$	$-\sin x$	$\sec^2 x$

The power rule $\frac{d}{dx}\,x^n = nx^{n-1}$ holds for *every* real exponent n, not just positive integers: it covers roots and reciprocals once they are written as powers, as in $\sqrt{x} = x^{1/2}$ and $1/x = x^{-1}$. (For a non-integer n this is read on $x > 0$, where x^n is defined.)

Example 2.1 ***(Reading the power rule both ways).***

$$\frac{d}{dx}\sqrt{x} = \frac{d}{dx}\,x^{1/2} = \tfrac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}, \qquad \frac{d}{dx}\,\frac{1}{x} = \frac{d}{dx}\,x^{-1} = -x^{-2} = -\frac{1}{x^2}.$$

Remark (Hyperbolic functions, briefly). The hyperbolic functions of Lecture 1 are differentiated by the same exponential rule, and we simply record the results:

$$\frac{\mathrm{d}}{\mathrm{d}x}\sinh x = \cosh x, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cosh x = \sinh x, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\tanh x = \operatorname{sech}^2 x.$$

Each follows in one line from $\sinh x = \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x})$ and $\cosh x = \frac{1}{2}(\mathrm{e}^x + \mathrm{e}^{-x})$. We will not dwell on them further.

2.2 The sum, product and quotient rules

Knowing the standard derivatives, we never return to the limit definition; instead we break a complicated function into simpler pieces and combine their derivatives with four rules. The first three handle sums, products and quotients.

Theorem 2.1 (Constant-multiple, sum, product and quotient rules). If u and v are differentiable and c is a constant, then so are cu, $u \pm v$, uv and (where $v \neq 0$) u/v, with

$$(cu)' = c\,u', \qquad (u \pm v)' = u' \pm v', \qquad (uv)' = u'v + uv', \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}.$$

Proof. Each follows from the limit definition. For the *sum*, the difference quotient of $u + v$ splits into the two separate quotients, whose limits are u' and v'. For the *product*, add and subtract a single cross term in the numerator:

$$\frac{u(x+\delta x)v(x+\delta x) - u(x)v(x)}{\delta x} = u(x+\delta x)\,\frac{v(x+\delta x) - v(x)}{\delta x} + \frac{u(x+\delta x) - u(x)}{\delta x}\,v(x).$$

As $\delta x \to 0$ the bracketed quotients tend to $v'(x)$ and $u'(x)$, while $u(x+\delta x) \to u(x)$ because a differentiable function is continuous (proved above); the limit is $uv' + u'v$. The constant-multiple rule is the special case $v = c$ of the product rule. The *quotient* rule follows from the same add-and-subtract trick applied to u/v, or more quickly by writing $u/v = u \cdot v^{-1}$ and using the chain rule of the next subsection. ∎

Note (The product rule is not $u'v'$). A common slip is to expect $(uv)' = u'v'$. It is not: differentiation does not distribute over a product. The correct rule keeps *two* terms, and a quick check confirms it—taking $u = v = x$ gives $(x^2)' = 2x$, whereas $u'v' = 1$.

Example 2.2 ***(Product rule).*** For $y = x^2 \sin x$, take $u = x^2$ and $v = \sin x$, so $u' = 2x$ and $v' = \cos x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = u'v + uv' = 2x\sin x + x^2\cos x = x\,(2\sin x + x\cos x).$$

Example 2.3 ***(Quotient rule, with*** $\tan x$ ***as a by-product).*** Writing $\tan x = \dfrac{\sin x}{\cos x}$ with $u = \sin x$, $v = \cos x$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x,$$

which is the entry tabulated above.

2.3 The chain rule

The most powerful rule differentiates a *composition*—a function of a function, such as $\sin(x^2)$ or e^{3x}. Recall from Lecture 1 that recognising the inner and outer functions is the key skill; the chain rule then multiplies their derivatives.

Theorem 2.2 (Chain rule). If $y = f(u)$ and $u = g(x)$, so that $y = f(g(x))$, then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x} = f'(g(x))\, g'(x).$$

In Leibniz notation the rule reads as though the $\mathrm{d}u$'s cancel—a useful mnemonic, though the $\mathrm{d}u$ are not literally numbers. In words: differentiate the outer function (keeping the inner one intact), then multiply by the derivative of the inner function.

Example 2.4 ***(A first chain rule).*** For $y = \sin(x^2)$, take the outer $f(u) = \sin u$ and inner $u = x^2$. Then $f'(u) = \cos u$ and $u' = 2x$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(x^2) \cdot 2x = 2x\cos(x^2).$$

A frequent special case is a function raised to a power, $y = \big(g(x)\big)^n$, for which the chain rule gives $y' = n\big(g(x)\big)^{n-1} g'(x)$.

Example 2.5 ***(Quotient then chain).*** Differentiate $y = \dfrac{(2x+1)^3}{\sqrt{x^2+1}}$. Use the quotient rule with $u = (2x+1)^3$ and $v = (x^2+1)^{1/2}$, each of which needs the chain rule:

$$u' = 3(2x+1)^2 \cdot 2 = 6(2x+1)^2, \qquad v' = \tfrac{1}{2}(x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}.$$

Then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{u'v - uv'}{v^2} = \frac{6(2x+1)^2\sqrt{x^2+1} - (2x+1)^3\dfrac{x}{\sqrt{x^2+1}}}{x^2+1}.$$

Multiplying through by $\sqrt{x^2+1}$ to clear the inner fraction and taking out the common factor $(2x+1)^2$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(2x+1)^2\big[6(x^2+1) - x(2x+1)\big]}{(x^2+1)^{3/2}} = \frac{(2x+1)^2\,(4x^2 - x + 6)}{(x^2+1)^{3/2}}.$$

Remark (Looking ahead: the sigmoid). The chain and quotient rules give the derivative of the logistic function $\sigma(x) = \big(1 + e^{-x}\big)^{-1}$ from Lecture 1. Differentiating,

$$\sigma'(x) = -\big(1 + e^{-x}\big)^{-2} \cdot \big(-e^{-x}\big) = \frac{e^{-x}}{(1 + e^{-x})^2} = \sigma(x)\big(1 - \sigma(x)\big),$$

the last step using $1 - \sigma(x) = \dfrac{e^{-x}}{1+e^{-x}}$. This tidy form—the derivative expressed in terms of the function itself—is what makes the sigmoid so convenient for training models, as the data-science students will see again in Lecture 8.

3 Further techniques

The rules so far assume y is given explicitly as $f(x)$. Three further techniques extend differentiation to functions described in other ways: through a parameter, implicitly by an equation, or as an awkward product whose logarithm is simpler.

3.1 Parametric differentiation

Sometimes a curve is traced out by giving both coordinates in terms of a third variable, the *parameter*:

$$x = x(t), \qquad y = y(t).$$

The circle $x = r\cos t,\ y = r\sin t$ and the ellipse $x = a\cos t,\ y = b\sin t$ are typical; so is the position of a particle, with t the time. To find the slope $\frac{dy}{dx}$ without eliminating t, apply the chain rule.

Theorem 3.1 (Parametric derivatives). If x and y are differentiable functions of a parameter t and $\dfrac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d^2y}{dx^2} = \frac{\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)}{dx/dt}.$$

The first formula is the chain rule $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ with $\frac{dt}{dx} = 1/(dx/dt)$. The second is the same rule applied again to the function $\frac{dy}{dx}$ of t—and note that it is *not* $(d^2y/dt^2)/(d^2x/dt^2)$, a tempting but wrong shortcut.

Example 3.1 ***(Slope of an ellipse).*** For $x = a\cos t,\ y = b\sin t$ we have $\dfrac{dx}{dt} = -a\sin t$ and $\dfrac{dy}{dt} = b\cos t$, so

$$\frac{dy}{dx} = \frac{b\cos t}{-a\sin t} = -\frac{b}{a}\cot t.$$

At $t = \frac{\pi}{2}$ (the top of the ellipse) the slope is 0, as expected.

Example 3.2 ***(Velocity components of a moving particle).*** A particle moves in the plane along $x = 3t^2 - 6t,\ y = 18t^2 - 2t^3$. The velocity components are the time derivatives $\dot{x} = 6t - 6$ and $\dot{y} = 36t - 6t^2$, and the path slope at time t is $\frac{dy}{dx} = \dfrac{36t - 6t^2}{6t - 6} = \dfrac{6t(6-t)}{6(t-1)} = \dfrac{t(6-t)}{t-1}$.

Example 3.3 ***(A parametric second derivative).*** For the curve $x = t^2$, $y = t^3$ (with $t \neq 0$), $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$ and $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 3t^2$, so the slope is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3t^2}{2t} = \tfrac{3}{2}t.$$

For the second derivative, differentiate this with respect to t and divide by $\mathrm{d}x/\mathrm{d}t$ again:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\dfrac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{3}{2}t\right)}{\mathrm{d}x/\mathrm{d}t} = \frac{3/2}{2t} = \frac{3}{4t}.$$

This is positive for $t > 0$ and negative for $t < 0$, so the curve is concave up on one branch and concave down on the other. The sign of $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is exactly the *concavity* test we will use for curve sketching in Lecture 8.

3.2 Implicit differentiation

Many curves are defined by an equation relating x and y that cannot be solved neatly for y, such as $x^2 + y^2 = 25$ or $\sin(x + y) = 3xy$. We can still differentiate the equation as it stands, treating y as an unknown function of x and applying the chain rule to every term involving y. The key move is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\, g(y) \,\right] = g'(y)\,\frac{\mathrm{d}y}{\mathrm{d}x},$$

so for instance $\frac{\mathrm{d}}{\mathrm{d}x}(y^2) = 2y\,\frac{\mathrm{d}y}{\mathrm{d}x}$ and, by the product rule, $\frac{\mathrm{d}}{\mathrm{d}x}(xy) = y + x\,\frac{\mathrm{d}y}{\mathrm{d}x}$. After differentiating, collect the $\frac{\mathrm{d}y}{\mathrm{d}x}$ terms and solve for it.

Example 3.4 ***(The circle $x^2 + y^2 = 25$).*** Differentiating both sides with respect to x,

$$2x + 2y\,\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \qquad\Longrightarrow\qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.$$

At the point $(3, 4)$ on the circle the slope is $-\frac{3}{4}$. Notice the answer involves both coordinates—exactly right for a curve that is not the graph of a single function.

Example 3.5 ***(An implicit second derivative).*** Let $x^2 + xy + y^2 = 12$, and write $y' = \frac{\mathrm{d}y}{\mathrm{d}x}$. Differentiating,

$$2x + \big(y + xy'\big) + 2yy' = 0 \qquad\Longrightarrow\qquad y' = -\frac{2x + y}{x + 2y}.$$

At the point $(-1, 2)$ this gives $y' = -\dfrac{-2+2}{-1+4} = 0$. To reach y'', differentiate the relation $2x + y + xy' + 2yy' = 0$ once more (product rule on xy' and $2yy'$):

$$2 + y' + \big(y' + xy''\big) + 2\big((y')^2 + yy''\big) = 0.$$

At $(-1, 2)$ with $y' = 0$ this reads $2 + 0 + (0 - y'') + 2(0 + 2y'') = 0$, i.e. $2 + 3y'' = 0$, so

$y'' = -\frac{2}{3}$. Being negative, it tells us the curve is concave down at that point—again the concavity test of Lecture 8, now for an implicit curve.

3.3 Logarithmic differentiation

When a function is a tall product or quotient of factors, or has the variable in both base *and* exponent (as in $y = x^x$), taking the natural logarithm first turns products into sums and exponents into multipliers, which are far easier to differentiate. The tool is the chain-rule fact

$$\frac{\mathrm{d}}{\mathrm{d}x} \ln|f(x)| = \frac{f'(x)}{f(x)},$$

the *logarithmic derivative* of f. Taking $\ln|f|$ rather than $\ln f$ matters: it makes the step legitimate wherever the factors are nonzero, whatever their sign, and the absolute values then cancel in the final answer. The recipe is: take $\ln|\cdot|$ of both sides, simplify with the log laws, differentiate implicitly, and multiply back by y.

Example 3.6 ***(A variable in the exponent: $y = x^x$).*** The power rule does not apply (x is in the base) and neither does the exponential rule (x is in the exponent). Take logs: $\ln y = x \ln x$. Differentiating the right-hand side by the product rule, and the left by the chain rule,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x + x \cdot \frac{1}{x} = \ln x + 1,$$

so $\frac{\mathrm{d}y}{\mathrm{d}x} = y\,(\ln x + 1) = x^x(\ln x + 1)$.

Example 3.7 ***(A four-factor quotient).*** For $y = \dfrac{\mathrm{e}^{-6x}\sin(4x)}{2x^2 - x + 1}$, logarithms break the quotient into a sum (the denominator $2x^2 - x + 1$ is always positive, so it needs no absolute value):

$$\ln|y| = -6x + \ln|\sin(4x)| - \ln(2x^2 - x + 1).$$

Differentiating term by term,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = -6 + \frac{4\cos(4x)}{\sin(4x)} - \frac{4x - 1}{2x^2 - x + 1} = -6 + 4\cot(4x) - \frac{4x - 1}{2x^2 - x + 1},$$

and multiplying back by y gives $\frac{\mathrm{d}y}{\mathrm{d}x}$ without ever applying the quotient rule to the original four-factor expression.

Summary

Concept	Key idea
Limit	$\lim_{x\to a} f(x) = L$: the value f approaches near a, ignoring $f(a)$. One-sided $\lim_{x\to a^\pm}$ (two-sided limit needs both equal); infinite limits give vertical asymptotes; limits at infinity give horizontal ones.
Continuity	f continuous at a if $\lim_{x\to a} f(x) = f(a)$; an unbroken graph. Polynomials, e^x, ln, sin, cos and their combinations are continuous.
Derivative	$f'(x) = \lim_{\delta x\to 0} \dfrac{f(x+\delta x) - f(x)}{\delta x}$, the slope of the tangent.
Three readings	Geometric (slope of tangent), physical (rate of change: velocity, acceleration), sensitivity (response to a nudge—the basis of gradient descent).
Notation	$f'(x)$, $\frac{dy}{dx}$, $\dot{x}$ (time); value at a written $f'(a)$ or $\left.\frac{dy}{dx}\right\rvert_{x=a}$.
Higher derivatives	f'', $f^{(n)}$; the second derivative measures how the slope changes (acceleration; concavity in Lecture 8).
Diff. vs continuity	Differentiable $\Rightarrow$ continuous, but not conversely: $\lvert x\rvert$ is continuous yet has a corner at 0 (left/right secant slopes ∓ 1).
Standard derivatives	$x^n \to nx^{n-1}$ (all real n), $e^{ax} \to ae^{ax}$, $\ln x \to \frac{1}{x}$, $\sin \to \cos$, $\cos \to -\sin$, $\tan \to \sec^2$.
Sum/product/quotient	$(u \pm v)' = u' \pm v'$; $(uv)' = u'v + uv'$; $(u/v)' = \dfrac{u'v - uv'}{v^2}$ (not $u'v'$).
Chain rule	$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$: outer derivative times inner derivative; powers $(g)^n \to ng^{n-1}g'$.
Parametric	$\frac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$; second derivative by differentiating this in t and dividing by dx/dt again.
Implicit	Differentiate the equation as is, using $\frac{d}{dx}g(y) = g'(y)\frac{dy}{dx}$, then solve for $\frac{dy}{dx}$ (and again for y'').
Logarithmic	Take ln first to tame products, quotients and powers like x^x; $\frac{d}{dx}\ln f = \dfrac{f'}{f}$.

Next lecture: differentiation put to work—tangents and normals, the key theorems and l'Hôpital's rule, classifying stationary points and sketching curves, and optimisation, including the idea of gradient descent.

Companion material: a separate *Exercise Sheet 7* (with solutions) develops the practice problems for this unit.